

Review 03 — Annular apertures (field-level Babinet)

mcdo_dev Phase 3 (output/03_annular/, docs/theory/03_annular_babinet.md)

2026-06-12

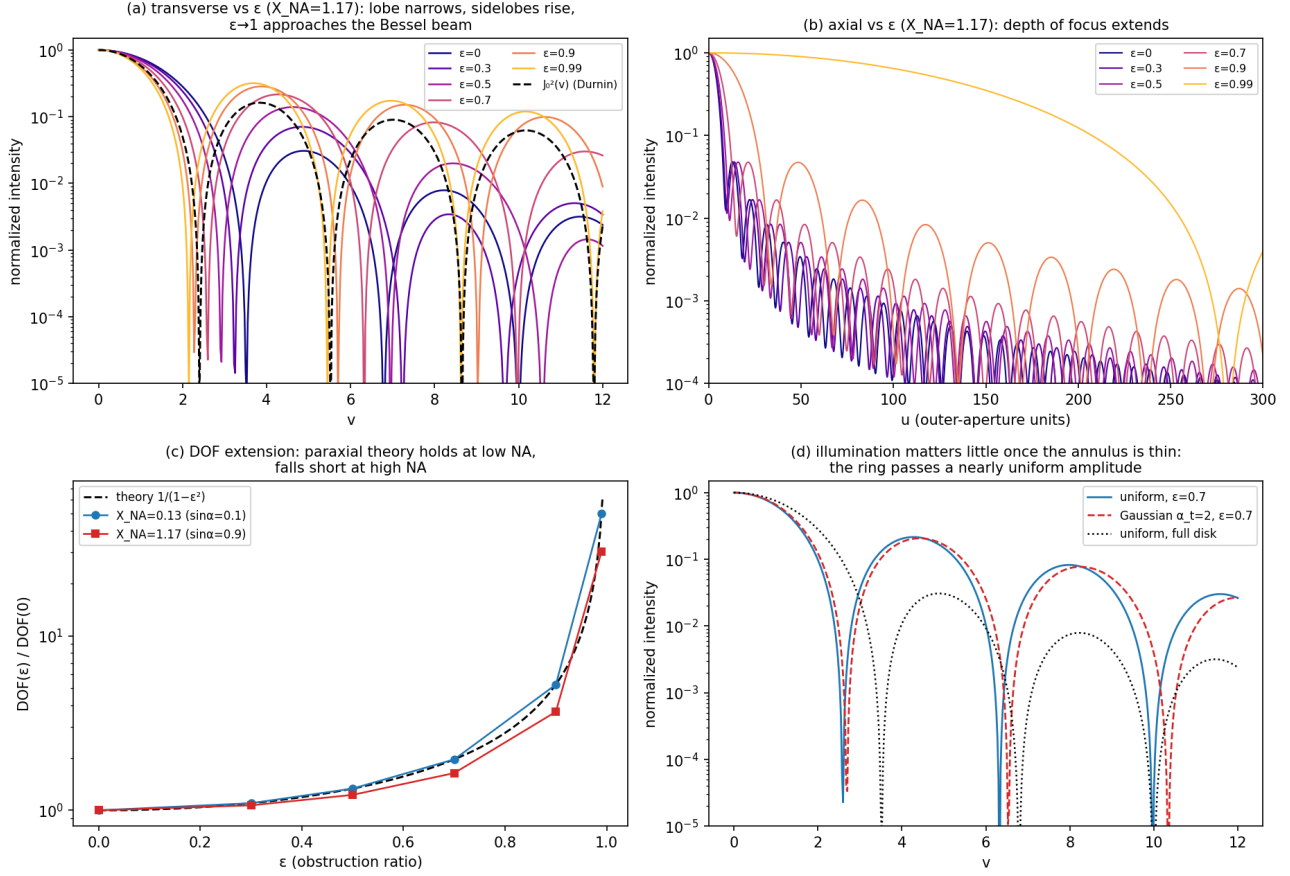
Construction and checkpoints

Annulus $\varepsilon = r_{in}/r_{out} = \sin \theta_{in}/\sin \alpha$ as the *field-level* difference of two disks at the same (r, z, k) : $I_m^{ann} = I_m(\alpha) - I_m(\alpha_{in})$, $\sin \alpha_{in} = \varepsilon \sin \alpha$; apodization referenced to the outer aperture (`mcdo/annular.py`). Linear- x input, y -cut.

- $\varepsilon = 0$ regression $\equiv$ full disk: difference exactly 0.
- **DOF extension matches paraxial theory to 3 digits at low NA** (50.13 vs $1/(1 - \varepsilon^2) = 50.25$ at $\varepsilon = 0.99$, $\sin \alpha = 0.1$); at $\sin \alpha = 0.9$ it falls $\sim 40\%$ short (30.4) at large ε .
- Central lobe narrows (FWHM_v 3.05 $\rightarrow$ 2.06 for ε 0 $\rightarrow$ 0.99 at $\sin \alpha = 0.9$) at the cost of strong sidelobes.
- **Thin-annulus limit:** $\rightarrow J_0^2(v)$ (Durnin) at low NA (max dev 0.004); at $\sin \alpha = 0.9$ the vector kernels deform the Bessel beam (max dev 0.163, FWHM 2.06 vs scalar 2.25) — input to Phase 4 (axicon).
- Illumination matters little once the annulus is thin: uniform vs Gaussian $\alpha_t=2$ at $\varepsilon = 0.7$: FWHM_v 2.450 vs 2.537.

ε	0.30	0.50	0.70	0.90	0.99
theory $1/(1 - \varepsilon^2)$	1.10	1.33	1.96	5.26	50.25
$X_{NA} = 0.13$	1.10	1.33	1.96	5.25	50.13
$X_{NA} = 1.17$	1.07	1.23	1.64	3.67	30.42

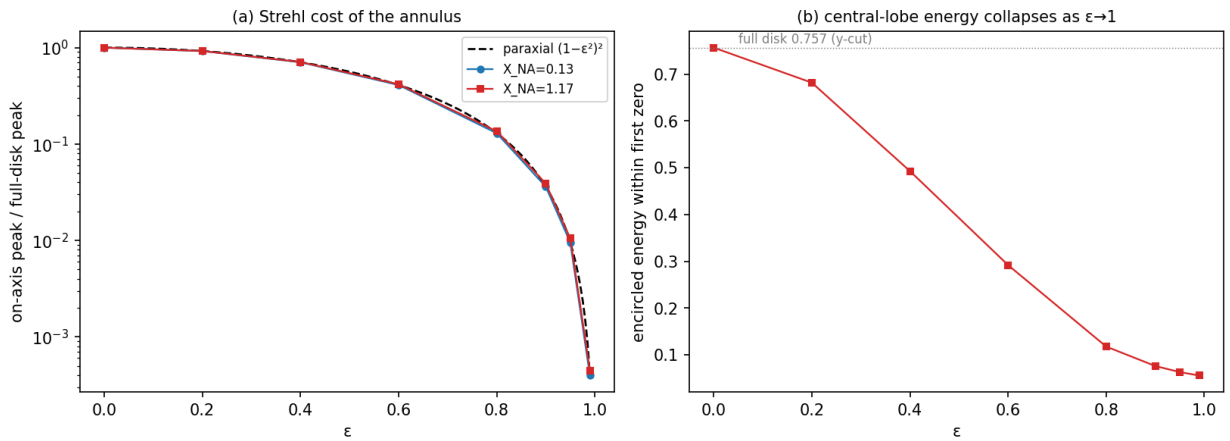
Phase 3 — annular apertures (field-level Babinet), linear-x y-cut



(a) transverse vs ϵ , approaching J_0^2 ; (b) axial extension; (c) DOF ratio vs theory at low/high NA; (d) uniform vs Gaussian for a thin annulus.

The cost side of the trade-off

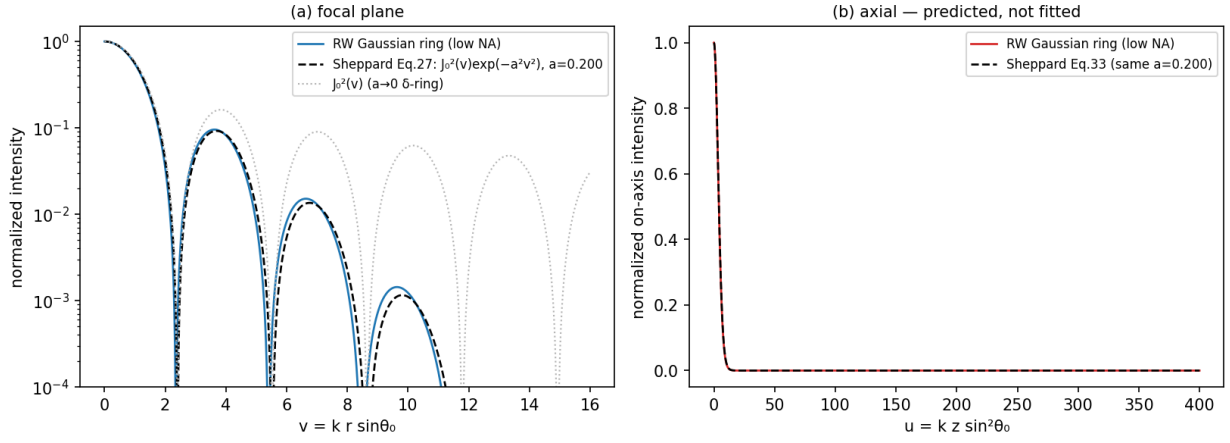
Phase 3 — the price of annular DOF extension (linear-x y-cut)



(a) on-axis Strehl tracks the paraxial $(1 - \epsilon^2)^2$ at both $X_{NA}=0.13$ and 1.17 (0.417 vs 0.410 at $\epsilon=0.6$; 4×10^{-4} at 0.99); (b) central-lobe encircled energy collapses $0.76 \rightarrow 0.055$ as $\epsilon \rightarrow 1$. The resolution/DOF gain is paid in peak intensity and contrast.

Literature anchor — Sheppard & Wilson 1978

Phase 3 anchor — Gaussian-ring pupil vs Sheppard & Wilson 1978 (low NA; a fitted from focal, predicts axial)



Their Laguerre-Gaussian mode sum gives closed forms for a Gaussian annulus (focal $J_0^2(v)e^{-v^2 a^2}$, axial $e^{-u^2 a^2/(1+u^2 a^4)}/(1+u^2 a^4)$). A Gaussian-ring pupil at low NA with $a = 0.200$ *fitted from the focal envelope* **predicts** the axial profile (Eq. 33, not fitted) to 0.8% (FWHM _{u} 8.2 vs 8.3) — an independent-method confirmation.